

RELEASE OF VIRUS AND BACTERIA FROM INFECTED CELLS: MODELS OF BUDDING AND BURSTING

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Abstract. When are virus or bacteria released from an infected cell? We consider two types of mathematical model. In the first, “budding”, infected cells are assumed to release new infectious particles at a constant rate (that is, constant probability per unit time). No description of the intracellular dynamics is needed; the mean number of new infectious particles released per infected cell is simply the rate of release multiplied by the mean lifetime of an infected cell. In the second, “bursting”, infectious particles accumulate inside a host cell until the cell dies and the intracellular load is released, all at once. At the level of an individual cell, the simplest stochastic budding models have two types of independent event: release of an infectious particle and death of the infected cell. In bursting, however, release of infectious particles and death of the host cell are not independent events: they occur simultaneously. We construct bursting models in which the death of an infected cell, releasing its cargo, is an event whose rate, like increase or decrease of the intracellular population, is proportional to the population size. Mean quantities describing release of new infectious particles in the bursting model satisfy simple differential equations that yield exact expressions for burst size distributions, and mean burst size conditioned on time of release. Even though the model has no imposed carrying capacity, but the mean intracellular population, given that the burst has not happened before time t , is an increasing function of time with a finite limiting value.

1. Introduction. Viruses, some bacteria, and bacteriophages use the environment inside host cells to reproduce, then exit and spread to other host cells. In the case of enveloped viruses [1], once a virus is assembled inside a host cell, its membrane can fuse with, and then pinch off from, the cell membrane. Thus, fully-formed virus particles can be released, one at a time, without cell lysis. On the other hand, exit of non-enveloped viruses is observed to occur in bursts, associated with lysis of the cell provoked by the presence of the virus [2]. This process has been observed in the case of bacteriophages since the early days of laboratory virology [3]. Some bacteria also reproduce inside cells. The bacterium *Francisella tularensis* is phagocytosed by alveolar macrophages, but is able to survive and escape from the phagosome to the cytosol [4, 5]. After multiple rounds of division in the cytosol, the bacterial load eventually causes the host macrophage to rupture and die, releasing many infectious bacteria [6, 7]. *B. anthracis* spores germinate in alveolar macrophages, replicate in the cytosol and are released when a cell ruptures and dies. These two schemes for the release of virus or bacteria, budding and bursting, require two different types of mathematical models [8]. Newly replicated and released particles may stand a better chance of survival if they move to other host cells *en masse*, than if they are released one at a time [9].

The number of new infectious particles released per infected cell is an important quantity in determining whether a small number of pathogenic agents inside a human or animal host will grow into a detectable infection. The number is an integer-valued random variable whose full distribution can only be calculated from the stochastic intracellular dynamics, even if the mean value may be determined from ordinary differential equations (ODEs). Similarly, reproduction numbers are integer-valued random variables whose distributions are determined by the stochastic dynamics of release and absorption [10].

A simplification used in most viral infection models is that the rate of release of new infectious particles is proportional to the number of infected cells [11–13]. That

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is, all infected cells are treated equally, regardless of how long they have been infected or how many infectious particles they hold. If V represents extracellular infectious particles and I represents infected cells, then the rate of change of V includes a positive term pI , where p is a constant. If the mean lifetime of an infected cell is $1/\delta$, then the mean number of new infectious particles released per infected cell is given by p/δ . Because of the linearity of these relationships, they can be taken to apply when V and I are interpreted as concentrations per unit volume, as the average over a cohort of cells, or as applying to a very small number of infected cells at the early stage of an infection. At the level of an individual cell, however, the dynamics is stochastic and the assumption behind such a “continuous production” model is that two types of event (release of an infectious particle and death of the infected cell) are independent. The events are not independent in models of bursting, where infectious particles accumulate inside a host cell until the cell dies and the whole intracellular load is released at once [8, 14].

At the stochastic level of an individual cell, the simplest budding model is that, in a time interval of duration Δt , an infected cell may release an infectious particle, with probability $p\Delta t$, or die, with probability $\delta\Delta t$. The budding and death rates, p and δ , do not depend on the number of infectious particles inside the cell, as long as there is at least one. The assumption can be modified by assuming that not all cells are productively infected, that infected cells pass through an “eclipse phase” before they begin to release infectious particles [15], or that infectious particles are released in cohorts [16–18]. We must modify the assumption in a more fundamental way in order to model bursting: the mechanism by which multiple infectious particles are released, all at once, when the infected cell ruptures and dies [19]. Release of infectious particles and death of infected cells are not independent processes. They are, rather, simultaneous.

Including intracellular dynamics, for example letting the rate of release of viral particles, or the cellular death rate, depend on the time a cell has been infected [20–23], tends to make a model more complicated. In one case, however, where the viral RNA inside each cell is governed by a linear ODE, it is possible to replace an age-structured model by an ODE [24, 25]. To construct a mathematical model of bursting, it may be assumed that the number of infectious particles released when a cell dies is a fixed parameter N [26], but the time at which the cell dies is a random variable. Alternatively, if the accumulation of infectious particles in an infected cell over time is explicitly modelled, and the time of cell death is an independent random variable, then the number of infectious particles released is also a random variable [10, 27].

In this work, we use an explicit stochastic model of the accumulation of infectious particles in an infected cell over time. Our bursting model of the intracellular population of virus or bacteria is a birth-and-death process with an extra type of event called “catastrophe” or “killing” where the population is eliminated [28–30]. The greater the load of infectious particles, the greater is the risk of the cell-death “catastrophe”. Each intracellular infectious particle has division rate λ . The model also contains a third type of event: death or removal of an intracellular infectious particle with rate μ , which makes it possible that a cell may cease to be infected altogether.

In the birth-death-catastrophe model of the intracellular population, the number of intracellular particles at time t is the integer-valued random variable $\mathbf{X}_t$. The random variables characterising the burst are

$$R, \quad \text{the burst size and} \quad \tau, \quad \text{the time of release.}$$

In the terminology of Karlin and Taylor [29], R is the “killing position”. Starting

with one intracellular infectious particle, the function $I(t)$ is the probability that the catastrophe *has not occurred* to that cell before t . We find simple expressions for I as a function of time and V as a function of I . The birth-catastrophe process exhibits interesting properties of quasi-stationarity when we consider the properties of $\mathbf{X}_t$ conditioned on the catastrophe not having occurred.

2. Budding. A class of within-host infection models [11–13, 23, 31–34] is based around the following pair of ODEs, used to calculate the basic reproduction number:

$$(2.1) \quad \begin{aligned} \frac{dI}{dt} &= \beta TV - \delta I \\ \frac{dV}{dt} &= pI - cV. \end{aligned}$$

The variables I and V represent infected cells and extracellular virus, respectively. The system of equations (2.1) may be augmented by an equation describing the dynamics of the host cells T or the pre-infection steady-state value of T may be used. Here, we are seeking to model the early stages of infection, when only a small number of cells are infected. To describe the dynamics inside a single infected cell, we take $I(0) = 1$ and $V(0) = 0$, in which case we would interpret $I(t)$ as “probability that the initially-infected cell is still alive and infected at time t ” rather than “concentration of infected cells at time t ”. With the goal of calculating how many viral particles are released from a single infected cell, regardless of their subsequent fate, the pair of ODEs is simplified to

$$(2.2) \quad \begin{aligned} \frac{dI}{dt} &= -\delta I \\ \frac{dV}{dt} &= pI. \end{aligned}$$

According to (2.2), if $V(0) = 0$ then $I(t) = I(0)e^{-\delta t}$ and $V(t) = \frac{p}{\delta}(1 - e^{-\delta t})I(0)$. To evaluate the mean number of infectious particles released from a single infected cell, we set $I(0) = 1$ and let $t \rightarrow \infty$:

$$V(t) \rightarrow \frac{p}{\delta}.$$

It is useful to combine (2.2) into

$$(2.3) \quad \frac{dV}{dI} = -\frac{p}{\delta}.$$

Thus, V and I are related by

$$(2.4) \quad V = \frac{p}{\delta}(1 - I).$$

In Figure 2.1, we plot I and V versus time, and use (2.4) to plot V versus I for this single-cell budding model.

When the differential equations (2.2) are interpreted as averages over many cells, they give no direct information about the intracellular dynamics. If we consider the stochastic dynamics of one cell at a time then the interpretation is that, in a short time interval Δt ,

1. an infected cell releases an infectious particle with probability $p\Delta t$;

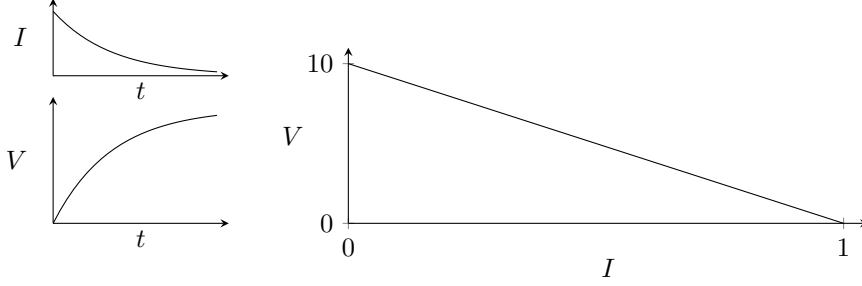


FIGURE 2.1. *Single-cell budding model.* The variables V and I represent mean quantities as a function of time since infection: $V(t)$ is the mean number of virus particles released from one infected cell, and $I(t)$ is the probability that a cell is still infected, up to time t . To model a single infected cell, $I(0) = 1$ and $V(0) = 0$. Here we show both quantities in the case of the budding model (2.2) with $p/\delta = 10$. Left panels: I and V as functions of time. Right: V versus I . (Because I is a decreasing function of time, we should imagine moving from right to left on the V - I graph.)

2. an infected cell dies with probability $\delta\Delta t$.

At any time in an infected cell, because two types of event are independent, the probability that the next event is release of a particle is equal to $p/(\delta + p)$. Pursuing this stochastic interpretation, the probability that an infected cell lives for a time greater than t is $I(t) = e^{-\delta t}$ and the number of infectious particles the cell releases in its lifetime is a random variable, R , with geometric distribution:

$$\begin{aligned} \mathbb{P}(R = k) &= \mathbb{P}(k \text{ release events before cell death}) \\ &= \frac{\delta}{\delta + p} \left(\frac{p}{\delta + p} \right)^k \quad k = 0, 1, 2, \dots \end{aligned}$$

3. Bursting. To model bursting, we construct the following explicit model of the intracellular population. The number of intracellular particles at time t is the discrete-valued random variable $\mathbf{X}_t$ that can take values $0, 1, 2, \dots$ and H . The notation $\mathbf{X}_t = H$ means that the cell has ruptured and released its intracellular content. Within the Markov process, the absorbing states 0 and H could be merged into one state, but we maintain the distinction for convenience when extending to models with natural cells death (from state 0) or reinfection of released particles.

If there are $\mathbf{X}_t$ intracellular particles at time t , and $(\lambda + \mu + \alpha)\Delta t \ll 1$, then one of three types of event may occur before $t + \Delta t$.

1. There may be one more particle at time $t + \Delta t$, with probability $\lambda\mathbf{X}_t\Delta t$.
2. There may be one fewer at time $t + \Delta t$, with probability $\mu\mathbf{X}_t\Delta t$.
3. The host cell may die before $\mathbf{X}_{t+\Delta t}$, releasing $\mathbf{X}_t$ infectious particles, with probability $\alpha\mathbf{X}_t\Delta t$.

We assume $\mu \geq 0$, $\alpha > 0$, $\lambda > 0$ and $\mathbf{X}_0 = 1$. Five individual realisations of the process thus defined are shown in Figure 3.1.

Let

$$p_n(t) = \mathbb{P}(\mathbf{X}_t = n | \mathbf{X}_0 = 1), \quad n = H, 0, 1, 2, \dots$$

Then

$$\frac{dp_H}{dt} = \alpha \sum_{n=1}^{\infty} np_n, \quad \frac{dp_0}{dt} = \mu p_1,$$

$p_1(0) = 1$ and, for $n \geq 1$,

$$\frac{dp_n}{dt} = \mu(n+1)p_{n+1} - (\alpha + \lambda + \mu)np_n + \lambda(n-1)p_{n-1}.$$

To analyse the process, we consider four functions of time. The first function is $I(t)$, the probability that the catastrophe *has not occurred* before t :

$$I(t) = \mathbb{P}(\mathbf{X}_t \neq H | \mathbf{X}_0 = 1) = 1 - p_H(t) = p_0(t) + p_1(t) + \dots$$

Thus $p_H(t) = 1 - I(t)$, which is the probability that the catastrophe *has occurred* before time t . An important property of $I(t)$ is [7, 30]

$$\mathbb{P}(\mathbf{X}_t \neq H | \mathbf{X}_0 = k) = (I(t))^k.$$

The second and third functions of time,

$$(3.1) \quad J(t) = p_1(t) + 2p_2(t) + 3p_3(t) + \dots \quad \text{and} \quad K(t) = p_1(t) + 4p_2(t) + 9p_3(t) + \dots,$$

have no analogue in the budding model. Because there is no intracellular population in state H , $J(t) = \mathbb{E}(\mathbf{X}_t | \mathbf{X}_0 = 1)$ and $K(t) = \mathbb{E}(\mathbf{X}_t^2 | \mathbf{X}_0 = 1)$. The fourth function is similar to that used in budding models: $V(t)$ is the mean number of infectious particles released up to t . The initial conditions are $I(0) = 1$, $J(0) = 1$, $K(0) = 1$ and $V(0) = 0$. The four functions $I(t)$, $J(t)$, $K(t)$ and $V(t)$ are smooth functions of time because they are means of time-dependent quantities.

In Figure 3.1, individual realisations of the bursting model are shown, along with the function $J(t)$ and the ratio $J(t)/I(t)$. Because the ultimate fate of any realisation is one of the absorbing states, $J(t) \rightarrow 0$ as $t \rightarrow \infty$ (dotted line). The dashed line, on the other hand, has a finite late-time limit. Because $J(t) = \mathbb{E}(\mathbf{X}_t | \mathbf{X}_t \neq H)I(t)$, the ratio $J(t)/I(t)$ is the mean population size inside cells that have not ruptured before time t .

We will use the function to analyse the random variables characterising bursting:

$$R, \quad \text{the burst size and} \quad \tau, \quad \text{the time of catastrophe.}$$

The probability that a burst of size n occurs between t and $t + \Delta t$ is $\alpha np_n(t)\Delta t$. The probability that $t < \tau < t + \Delta t$ is the probability that a burst of any size occurs between t and $t + \Delta t$, which is

$$(3.2) \quad \mathbb{P}(\text{cell death in } (t, t + \Delta t)) = \alpha(p_1(t) + 2p_2(t) + 3p_3(t) + \dots)\Delta t = \alpha J(t)\Delta t.$$

Similarly,

$$(3.3) \quad \begin{aligned} \text{mean number of particles released in } (t, t + \Delta t) = \\ \alpha(p_1(t) \times 1 + 2p_2(t) \times 2 + 3p_3(t) \times 3 + \dots)\Delta t = \alpha K(t)\Delta t. \end{aligned}$$

Using (3.2) and (3.3), we write three differential equations

$$(3.4) \quad \begin{aligned} \frac{dI}{dt} &= -\alpha J \\ \frac{dJ}{dt} &= (\lambda - \mu)J - \alpha K \\ \frac{dV}{dt} &= \alpha K. \end{aligned}$$

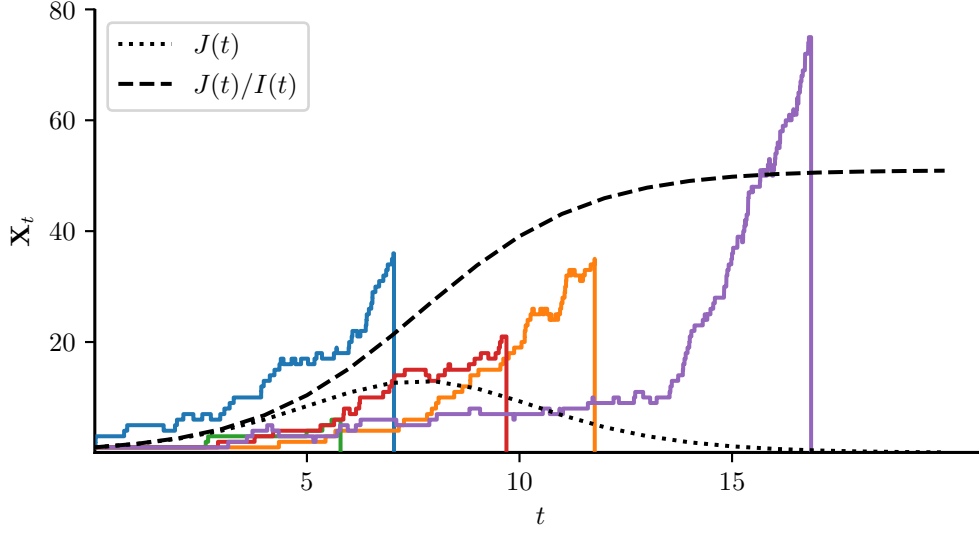


FIGURE 3.1. Realizations of the birth-death process with catastrophe: bursting model of the intracellular dynamics of a single infected cell. The coloured lines are five individual realisations with $\mathbf{X}_0 = 1$ and parameter values $\lambda = 0.5$, $\mu = 0.1$ and $\alpha = 0.01$. The dotted line is the mean value of the population at time t , which is a smooth function of time that tends to zero as $t \rightarrow \infty$. The dashed line, with finite limiting value reminiscent of a carrying capacity, is the conditional mean value of the population at time t , given that the catastrophe event has not yet occurred.

These equations are more complicated than their budding-model equivalent (2.2). Nevertheless, we can find an analytical solution.

Using $\alpha K = (\lambda - \mu)J - \frac{dJ}{dt}$, we find an expression for V :

$$\begin{aligned}
 V(t) &= (\lambda - \mu) \int_0^t J(s) ds - \int_0^t \frac{dJ}{ds} ds \\
 &= -\frac{\lambda - \mu}{\alpha} \int_0^t \frac{dI}{ds} ds - \int_0^t \frac{dJ}{ds} ds \\
 (3.5) \quad &= \frac{\lambda - \mu}{\alpha} (I(0) - I(t)) + J(0) - J(t) \\
 &= \frac{\lambda - \mu}{\alpha} (1 - I(t)) + 1 - J(t).
 \end{aligned}$$

The solution (3.5) does not yet represent a solution of the *complete set* of ODEs (3.4). Such a solution is obtained if we also use the result of Gillard *et al.*, who showed that [7, 30] I satisfies the closed differential equation

$$(3.6) \quad \frac{dI}{dt} = \mu - (\lambda + \alpha + \mu)I + \lambda I^2.$$

(A short derivation is given in Appendix A.) We may rewrite (3.6), using (3.4), as

$$(3.7) \quad \alpha J = -\mu + (\lambda + \alpha + \mu)I - \lambda I^2.$$

We differentiate (3.7) and use (3.4) to obtain an expression for K :

$$(3.8) \quad \alpha K = (2\lambda + \alpha - 2\lambda I)J.$$

Thus, in Figure 3.2, we plot J and K as functions of I . In both cases, they depend on the parameters via the ratios λ/α and μ/α .

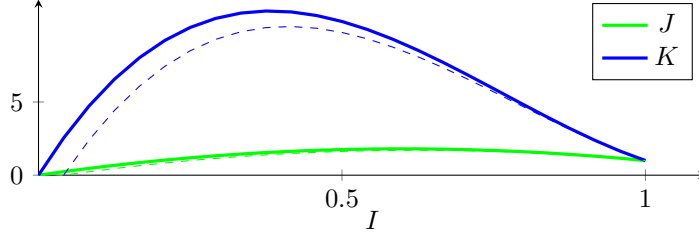


FIGURE 3.2. J and K as functions of I , using (3.7) and (3.8). Recall that the direction of movement as time increases is from right to left. Solid lines: $\lambda/\alpha = 5$ and $\mu = 0$. As $t \rightarrow \infty$ (moving leftwards in the Figure), $I(t) \rightarrow 0$ and $J(t) \rightarrow 0$. Dashed lines: $\lambda/\alpha = 5$ and $\mu/\alpha = 0.25$. As $t \rightarrow \infty$, $I(t)$ has a finite limit while $J(t) \rightarrow 0$.

Our four functions also have interpretations via probabilities and conditional probabilities of the random variables τ and R . The mean burst size is $\mathbb{E}(R) = \lim_{t \rightarrow \infty} V(t)$. Burst-size probabilities, conditioned on τ , are

$$(3.9) \quad \mathbb{P}(R = n | \tau = t) = \frac{np_n(t)}{J(t)} \quad n = 1, 2, 3, \dots$$

Given that rupture occurs at time t , the mean burst size is given by

$$(3.10) \quad \mathbb{E}(R | \tau = t) = \sum_n n \mathbb{P}(R = n | \tau = t) = \frac{K(t)}{J(t)}.$$

3.1. Bursting dynamics without intracellular death. In the case $\mu = 0$, the following two parameter combinations are convenient:

$$r = \frac{\lambda}{\alpha} \quad \text{and} \quad \sigma = \lambda + \alpha.$$

The solution of (3.6) is written

$$(3.11) \quad I(t) = \frac{r+1}{r + e^{\sigma t}}.$$

We express $J(t)$ and $V(t)$ in terms of $I(t)$, with $\mu = 0$, by simplifying (3.7) and (3.5):

$$(3.12) \quad J = (r+1)I - rI^2 \quad \text{and} \quad V = r+1 - (2r+1)I + rI^2.$$

As $t \rightarrow \infty$,

$$(3.13) \quad I(t) \rightarrow 0, \quad \frac{J(t)}{I(t)} \rightarrow r+1 \quad \text{and} \quad V(t) \rightarrow r+1.$$

In Figure 3.3, we construct the equivalent of Figure 2.1, comparing V and I as functions of time.

It is interesting to compare the relationship $\frac{dV}{dt} = \alpha K$ (bursting) with the relationship $\frac{dV}{dt} = pI$ (budding). We may write

$$(3.14) \quad \frac{dV}{dt} = \alpha \frac{K}{I} I = \alpha(2r+1 - 2rI(t))(r+1 - rI(t))I.$$

As $t \rightarrow \infty$, $\frac{K(t)}{I(t)} \rightarrow (2r+1)(r+1)$. Thus, at late times, the bursting model does exhibit the property that $\frac{dV}{dt}$ is proportional to I .

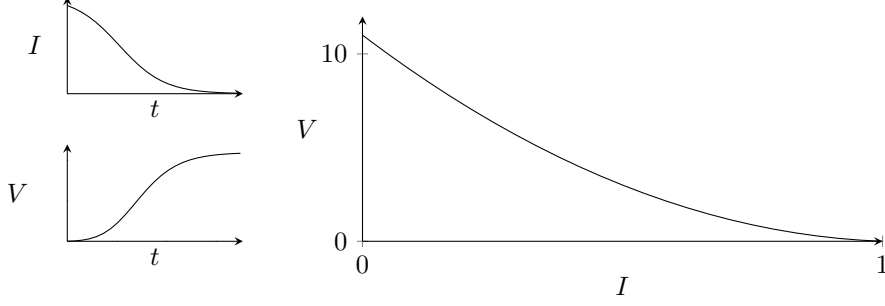


FIGURE 3.3. *Single-cell bursting model. Left panels: I and V as functions of time using (3.11) and (3.12). Right: V versus I using (3.12). Here $\lambda/\alpha = 10$ and $\mu = 0$. The corresponding graphs in the budding model are in Figure (2.1).*

The probability density of τ is given by $-I'(t)$. Using integration by parts and (3.11),

$$(3.15) \quad \mathbb{E}(\tau) = - \int_0^\infty t I'(t) dt = \int_0^\infty I(t) dt = \frac{1}{\lambda} \log(r+1),$$

For a comparison of our budding with bursting models, in the case $\mu = 0$, see Table 3.1.

	budding	bursting
$\frac{dI}{dt}$	$-\delta I$	$-(\lambda + \alpha)I + \lambda I^2$
$I(t)$	$e^{-\delta t}$	$\frac{\lambda + \alpha}{\lambda + \alpha e^{(\lambda + \alpha)t}}$
mean cell lifetime	$1/\delta$	$\frac{1}{\lambda} \log(1 + \lambda/\alpha)$
V	$\frac{p}{\delta}(1 - I)$	$\frac{\lambda}{\alpha} + 1 - \frac{2\lambda + \alpha}{\alpha}I + \frac{\lambda}{\alpha}I^2$

TABLE 3.1

Summary of important relationships between V and I when $\mu = 0$. Budding parameters: p is rate of release of particles from, and δ is death rate of, an infected cell.

Bursting parameters: λ is division (birth) rate of intracellular particles and α is rupture (catastrophe) rate, per intracellular particle, of an infected cell.

If $\mu = 0$ then $p_0(t) = 0$ at any t and $p_H(t) \rightarrow 1$ as $t \rightarrow \infty$. In other words, the catastrophe happens, sooner or later, in every realisation. Until the catastrophe occurs, the distribution of $\mathbf{X}_t$ at any time is geometric [29, 30]. That is, there is a function $g(t)$ such that

$$(3.16) \quad p_k(t) = I(t)g(t)(1 - g(t))^{k-1} \quad k = 1, 2, \dots$$

The mean intracellular population size at time t , given that the catastrophe has not yet occurred, is equal to $1/g(t)$. Using the definition (3.1),

$$(3.17) \quad J(t) = \frac{I(t)}{g(t)} \quad \text{and} \quad K(t) = I(t) \left(\frac{2}{g^2(t)} - \frac{1}{g(t)} \right).$$

Using (3.11) and (3.12), we write

$$(3.18) \quad g(t) = \frac{I(t)}{J(t)} = \frac{re^{-\sigma t} + 1}{r + 1}.$$

Note that $g(t)$ is a decreasing function of time with $g(0) = 1$ and, as $t \rightarrow \infty$, $g(t) \rightarrow \frac{1}{r+1}$. That is, we find the somewhat counter-intuitive results that the mean population size, given that the catastrophe has not occurred, tends to the constant $r+1$ as $t \rightarrow \infty$.

Still with $\mu = 0$, we may write

$$(3.19) \quad p_n(t) = e^{-\sigma t} \left(\frac{1 - e^{-\sigma t}}{1 + 1/r} \right)^{n-1} \quad n = 1, 2, \dots$$

Given n , $p_n(t)$ is a function of time that takes its maximum value when $t = t_n$ where

$$(3.20) \quad t_n = \frac{\log n}{\sigma}.$$

The mean value of rupture time, for a burst of size n , is equal to $t_n + \gamma/\sigma$, where $\gamma = 0.57 \dots$ (Euler's constant). Two examples are shown in Figure 3.5.

The probability that a burst of size n happens between t and $t + \Delta t$ is $\alpha n p_n(t) \Delta t$. Therefore, the distribution of burst sizes is given by [7, 30]

$$(3.21) \quad \begin{aligned} \mathbb{P}(R = n) &= \alpha n \int_0^\infty p_n(t) dt \\ &= \alpha n \left(\frac{r}{r+1} \right)^{n-1} \frac{1}{\sigma} \int_0^\infty e^{-s} (1 - e^{-s})^{n-1} ds \\ &= \frac{1}{r} \left(\frac{r}{r+1} \right)^n \quad n = 1, 2, \dots \end{aligned}$$

This last result can be compared to a histogram constructed from an experiment that measures burst sizes. Further comparisons, of conditional quantities, are possible if experimental measurement of rupture times is also made. For example, Figure 3.4 shows the mean burst size conditioned on rupture time, using (3.10) and (3.17)-(3.18). Each experimentally-measured burst size-burst time pair can be represented as a dot on the Figure. Note the late-time limit of the mean burst size: $2r - 1$. The right-hand panel of Figure 3.5 shows the mean rupture time, conditioned on rupture size. With (3.19), the exact expression is (see Appendix B)

$$(3.22) \quad \mathbb{E}(\tau | R = n) = \frac{\int_0^\infty t p_n(t) dt}{\int_0^\infty p_n(t) dt} = \frac{1}{\sigma} \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \right).$$

A useful large- n approximation to (3.22) $\mathbb{E}(\tau | R = n) \simeq \frac{1}{\sigma} (\ln n + \gamma)$. In the same large- n limit, the distribution of the conditioned rupture time has a Gumbel form,

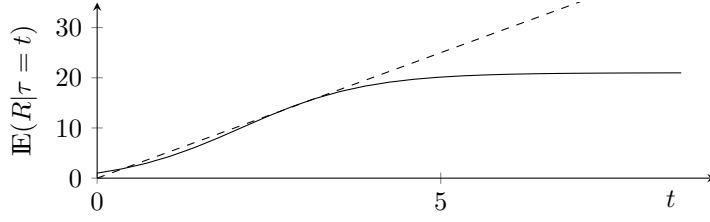


FIGURE 3.4. The mean burst size is an increasing function of the time of rupture. Solid line: the dependence of conditional mean burst size, $\mathbb{E}(R|\tau = t)$, on t (3.10). With $\mu = 0$, $K(t)/J(t) = 2/g(t) - 1$. The plot uses (3.17) and (3.18) with $r = 10$ and $\sigma = 1.1$. The dashed line is the corresponding quantity in the budding model, $\mathbb{E}(R|\tau = t) = pt$, with $p = 5$.

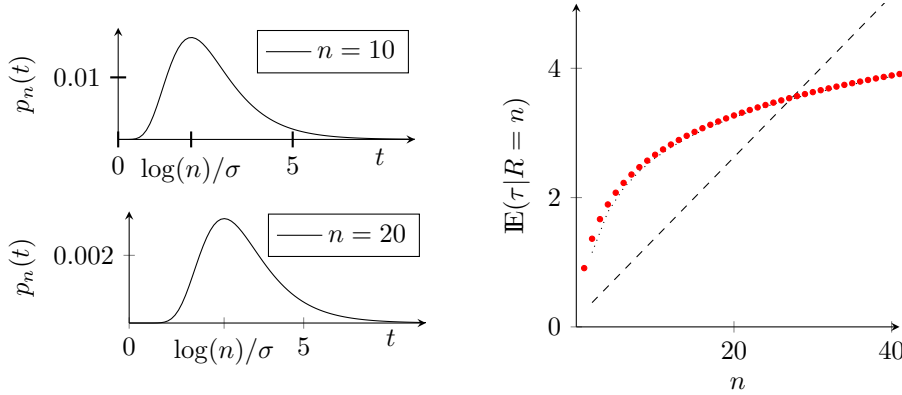


FIGURE 3.5. Left panels: The probability that there are n intracellular particles, as a function of time, with $n = 10$ and $n = 20$. (The choice of n changes the time the maximum is attained, but has little effect on the shape of the curve.) Right panel: Conditional mean time, given rupture size n . The red dots show the exact mean rupture time as a function of rupture size, in the bursting model with $\mu = 0$ and $\lambda + \alpha = 1.1$. The dotted line is the large- n approximation $(\ln n + \gamma)/(\lambda + \alpha)$. The dashed straight line is the corresponding budding expression, with $\delta + p = 8$.

illustrated on the left-hand side in Figure 3.5:

$$\begin{aligned}
 \mathbb{P}(\tau < t | R = n) &= \frac{\alpha n \int_0^t p_n(s) ds}{\mathbb{P}(R = n)} \\
 (3.23) \quad &= (1 - e^{-\sigma t})^n \\
 &\simeq \exp(-ne^{-\sigma t}) \quad \text{if } n \gg 1 \\
 &= \exp(-e^{-T}) \quad \text{where } T = \sigma(t - t_n).
 \end{aligned}$$

The dashed line on the right-hand side in Figure 3.5 is the expression corresponding to (3.22), assuming budding, for the mean time to cell death, with release of exactly n particles. Then, because there are $n + 1$ events, each with the same mean waiting time, the conditional mean time is equal to $(n + 1)/(\delta + p)$.

3.2. The effect of death of infectious particles. We now relax the assumption that $\mu = 0$. Events in which an intracellular pathogen is eliminated by the cell are allowed, so that a cell may eliminate its intracellular population, avoiding lysis. That is, if $\mu > 0$ then $p_0(t) > 0$ when $t > 0$. The probability that a cell is still alive and infected at time t , is equal to $I(t) - p_0(t)$.

Let us return to the differential equation (3.6) for the infected cell survival probability, written as [7, 30]. Because the RHS is a quadratic function of I , we can write

$$(3.24) \quad \frac{dI}{dt} = -\lambda(b - I)(I - a).$$

If λ , α and μ are all greater than zero then $0 < a < 1 < b$ and $ab = \mu/\lambda$. We may write $J(t) = \frac{\lambda}{\alpha}(b - I(t))(I(t) - a)$ and, using (3.8),

$$\frac{dV}{dt} = (I - a) \frac{\alpha K}{I - 1} = \frac{\lambda}{\alpha}(I - a)(2\lambda(1 - I) + \alpha)(b - I).$$

As $t \rightarrow \infty$, $I(t) \rightarrow a$, $J(t) \rightarrow 0$, $K(t) \rightarrow 0$ and $\frac{dV}{dt}$ is proportional to $I - a$. Generalising (3.13), the mean total burst size is obtained from (3.5):

$$(3.25) \quad \lim_{t \rightarrow \infty} V(t) = \frac{\lambda - \mu}{\alpha}(1 - a) + 1.$$

Recall that the quantity a represents the probability that the cell eliminates the pathogen, avoiding lysis. Amongst cells that do release pathogen, the mean burst size is equal to $\frac{\lambda - \mu}{\alpha} + 1/(1 - a)$.

With $\mu > 0$, the $p_k(t)$ are expressed in terms of the following function:

$$h(t) = \frac{1 - e^{-\lambda(b-a)t}}{b - ae^{-\lambda(b-a)t}}.$$

Note that $h(0) = 0$ and $h(t) \rightarrow \frac{1}{b}$ as $t \rightarrow \infty$. We write $p_0(t) = \frac{\mu}{\lambda}h(t)$ and [30]

$$p_k(t) = \left(\frac{b - a}{b - ae^{-\lambda(b-a)t}} \right)^2 e^{-\lambda(b-a)t} (h(t))^{k-1} \quad k \geq 1.$$

The burst size is zero (the cell survives) with finite probability: $\mathbb{P}(R = 0) = a$. For non-zero burst sizes [29, 30],

$$\mathbb{P}(R = n) = (1 - a)(b - 1) \frac{1}{b^n} = \frac{\alpha}{\lambda} \frac{1}{b^n} \quad n \geq 1.$$

Note the geometric form of the distribution, generalising (3.21).

When $\mu > 0$, the burst time τ does not have finite mean. However, we may consider conditional means:

$$(3.26) \quad \mathbb{E}(\tau | \tau < \infty) = \frac{1}{1 - a} \int_0^\infty (I(t) - a) dt = \frac{1}{\lambda(1 - a)} \log \left(\frac{b - a}{b - 1} \right),$$

which is the generalisation of (3.15). Properties of the random variables R and τ , are summarised in Table 3.2, with $\mu = 0$ and $\mu > 0$ (right).

A fairly simple expression for $\mathbb{E}(\tau | R = n)$ exists in the limit $\frac{\mu}{\lambda} \ll 1$ and $\frac{\alpha}{\lambda} \ll 1$. Then $a \simeq \frac{\mu}{\lambda}$, $b \simeq 1 + \frac{\alpha}{\lambda}$ and

$$h(t) \simeq 1 - e^{-(\sigma - \mu)t} \left(1 - \frac{\mu}{\lambda} \right).$$

Consequently, $p_k(t)$ takes its maximum value at $t = t_{\max}$ where $(\sigma - \mu)t_{\max} \simeq \log k - \frac{\mu}{\lambda}$. Thus

$$(3.27) \quad \mathbb{E}(\tau | R = n) \simeq (\log n + \gamma - \frac{\mu}{\lambda}) / (\sigma - \mu).$$

In Figure 3.6, we show the conditional mean rupture time as a function of rupture size, given three values of μ . The dashed lines, barely visible below the numerical results, use (3.27).

	$\mu = 0$	$\mu > 0$
$\mathbb{E}(R)$	$\frac{\lambda}{\alpha} + 1$	$\frac{\lambda - \mu}{\alpha}(1 - a) + 1$
$\mathbb{P}(R = n), n = 1, 2, \dots$	$\frac{\alpha}{\lambda} \left(\frac{\lambda + \alpha}{\alpha} \right)^n$	$\frac{\alpha}{\lambda} \frac{1}{b^n}$
$\mathbb{E}(\tau \tau < \infty)$	$\frac{1}{\lambda} \log\left(\frac{\lambda}{\alpha} + 1\right)$	$\frac{1}{\lambda(1 - a)} \log\left(\frac{b - a}{b - 1}\right)$
$\lim_{t \rightarrow \infty} \mathbb{E}(R \tau = t)$	$2\frac{\lambda}{\alpha} + 1$	$2\frac{\lambda}{\alpha}(1 - a) + 1$

TABLE 3.2

Bursting: the time of release, τ , and the burst size R . The top two lines show the mean burst size and distribution of burst sizes, with ($\mu > 0$) and without ($\mu = 0$) intracellular death. With probability equal to a , a cell's intracellular population is eliminated without any release. The mean rupture time (third line) is conditioned on $R > 0$. The fourth line is the mean burst size, as a function of τ . The quantities a and b are the roots of the quadratic equation $\lambda I^2 - (\lambda + \mu + \alpha)I + \mu = 0$.

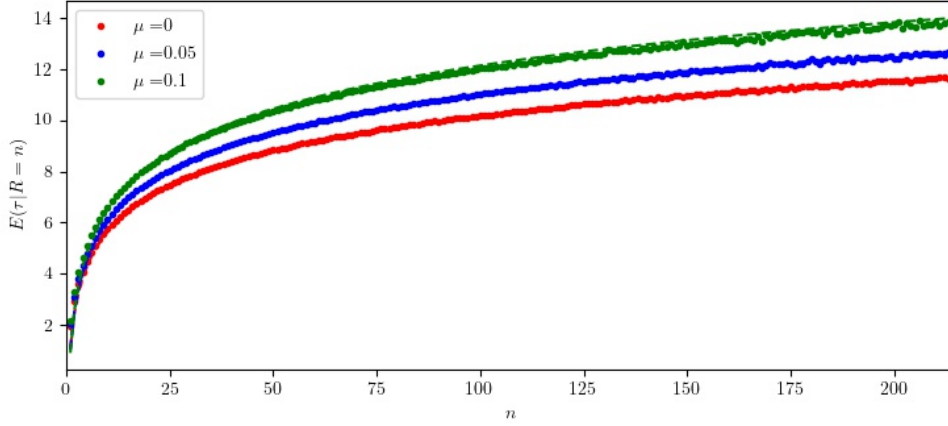


FIGURE 3.6. The mean burst time conditioned on burst size n , with three values of μ . The dashed lines use (3.27). $\lambda = 0.5$, $\alpha = 0.01$

4. Discussion. The number of new infectious particles released per infected cell [10] is an integer-valued random variable that is key to determining whether a small number of pathogenic agents inside a human or animal host will grow into a detectable infection. It may be possible to calculate the mean value using ODEs with simple assumptions, but calculating the full distribution requires considering the intracellular dynamics. Here, we compare two models: budding and bursting. Both types of models use the variables, $V(t)$ and $I(t)$, that represent free virus and infected cells. To model bursting, we use an explicit stochastic model of the accumulation of infectious particles in an infected cell over time. The model is a birth-and-death process with an extra type of event called “catastrophe” [28–30]. Despite the explicit stochastic representation of intracellular dynamics, and increased number of variables compared to the budding model, we find simple analytical expressions for burst-size distributions, means of time-dependent quantities and conditional probabilities.

Data obtained when mice were challenged with *Francisella tularensis* SCHU S4, yielded estimated λ , μ and α values of 0.15 h^{-1} , 0.01 h^{-1} , and 0.00015 h^{-1} [30]. The predicted fraction of cells that do not release any bacteria, equal to the quantity a in (3.24), is less than 7 percent, and the distribution of burst sizes from cells that rupture was predicted to be geometric with typical burst sizes of hundreds of bacteria [7, 35]. A consequence is that a dose as low as 10 CFU of *Francisella tularensis* can result in infection [36]. Recently, Rawsthorne-Manning *et al.* (private communication; manuscript in preparation), observed J774 cells heavily infected with *Francisella tularensis*. Video recording of bursting events with *en masse* bacterial release, and use of high-magnification confocal microscopy to count bacterial payload within cells that were adjudged to be close to rupturing, provide observational support to the burst-size prediction.

Very different parameter estimates are found for *Bacillus anthracis*. Approximate Bayesian Computation Sequential Monte Carlo using published experimental time-series data produced the estimates $\lambda \approx 0.64 \text{ h}^{-1}$, $\mu \approx 1.64 \text{ h}^{-1}$, and $\alpha \approx 0.04 \text{ h}^{-1}$ [37]. Consequently, more than 96percent of phagocytes are predicted to eliminate an intracellular *Bacillus anthracis* infection. Using (3.25), the remaining fraction of phagocytes that do rupture are predicted to release only an average of 1.6 bacteria each, consistent with estimates that the dose required for infection is between 8×10^3 and 5×10^4 spores. The dynamics of *coxiella burnetti*, which adopts a small form outside a host cell and a large form inside, may be similar. Stochastic population dynamics has been used to model the highly variable timing and magnitude of HIV release from reactivated memory CD4 T cells from infected donors [14]: on some days more than 10000 HIV RNA copies were found in single-cell well supernatant, on other days nothing was detected.

Influenza A [?, 19]

The budding model is consistent with the assumption is that every infected cell behaves independently, selecting one of two possible events (virion release or cell death) in each time interval. In bursting, on the other hand, infectious particles accumulate inside a host cell until the cell dies and the intracellular load is released at once. Then, release of infectious particles and death of the host cell are not independent events, but occur simultaneously. Our two scenarios of budding and bursting can be thought of as opposite ends of a spectrum of possibilities. Intermediate scenarios to consider include internal compartments that provide mechanisms for release of multiple infectious particles without immediate lysis of the host cell [38]. Such vesicles may enhance infectivity by allowing multiple viral genomes to enter a nearby cell [39]. A related challenge is producing models of viruses, including plant viruses, that are transmitted

directly from cell to cell and not released into the extracellular environment.

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Appendix A. The survival function and the differential equation (3.6).

An important consequence of the linear form of the rates in Section 3 is that

$$\mathbb{P}(\text{infected cell still alive at } t | \mathbf{X}_0 = k) = (I(t))^k.$$

Suppose $\mathbf{X}_0 = 1$ and consider a time interval Δt such that $(\lambda + \alpha + \mu)\Delta t \ll 1$. Then

$$\begin{aligned}\mathbb{P}(\mathbf{X}_{\Delta t} = 0) &= \mu\Delta t + \mathcal{O}(\Delta t^2) \\ \mathbb{P}(\mathbf{X}_{\Delta t} = H) &= \alpha\Delta t + \mathcal{O}(\Delta t^2) \\ \mathbb{P}(\mathbf{X}_{\Delta t} = 2) &= \lambda\Delta t + \mathcal{O}(\Delta t^2).\end{aligned}$$

Next, consider $I(t + \Delta t) = \mathbb{P}(\mathbf{X}_{t+\Delta t} \neq H)$ via the three possibilities at $t = \Delta t$:

$$\begin{cases} \text{If } \mathbf{X}_{\Delta t} = 0 \text{ then } \mathbf{X}_{t+\Delta t} = 0 \\ \text{If } \mathbf{X}_{\Delta t} = H \text{ then } \mathbf{X}_{t+\Delta t} = H \\ \text{If } \mathbf{X}_{\Delta t} = 2 \text{ then } \mathbb{P}(\mathbf{X}_{t+\Delta t} \neq H) = I(t)^2. \end{cases}$$

Thus [7, 30]

$$(A.1) \quad I(t + \Delta t) = (1 - (\lambda + \alpha + \mu)\Delta t)I(t) + \mu\Delta t + \lambda\Delta t I(t)^2.$$

In other words

$$\frac{I(t + \Delta t) - I(t)}{\Delta t} = \mu - (\lambda + \alpha + \mu)I(t) + \lambda I(t)^2.$$

The solution of (3.6), with $I(0) = 1$, is

$$I(t) = \frac{a(b-1) + b(1-a)e^{-\lambda(b-a)t}}{b-1 + (1-a)e^{-\lambda(b-a)t}},$$

where

$$a, b = \frac{1}{2\lambda} \left(\lambda + \alpha + \mu \mp ((\lambda + \alpha + \mu)^2 - 4\lambda\mu)^{1/2} \right).$$

We may also add a fourth type of event: natural death of uninfected cells with rate δ . Then $p_k(t)$ is unaffected for $k > 0$ but

$$\begin{aligned}\frac{dp_0}{dt} &= \mu p_1 - \delta p_0 \\ \frac{dp_H}{dt} &= \alpha \sum_{n=1}^{\infty} n p_n + \delta p_0(t).\end{aligned}$$

Appendix B. Integrals used to calculate mean rupture time.

Let

$$L_1(n) = \int_0^{\infty} e^{-s}(1 - e^{-s})^{n-1} ds \quad L_2(n) = \int_0^{\infty} s e^{-s}(1 - e^{-s})^{n-1} ds,$$

$n = 1, 2, 3, \dots$. Then $L_1(1) = 1$ and $L_2(1) = 1$.

Firstly

$$L_1(n) = \frac{1}{n} [(1 - e^{-s})^n]_0^{\infty} = \frac{1}{n}.$$

Now

$$\begin{aligned}
L_2(n+1) &= \int_0^\infty se^{-s}(1-e^{-s})^n ds \\
&= \int_0^\infty se^{-s}(1-e^{-s})(1-e^{-s})^{n-1} ds \\
&= L_2(n) - \int_0^\infty se^{-2s}(1-e^{-s})^{n-1} ds \\
&= L_2(n) + \frac{1}{n} \int_0^\infty (e^{-s} - se^{-s})(1-e^{-s})^n ds
\end{aligned}$$

(integration by parts)

$$= L_2(n) + \frac{1}{n} L_1(n+1) - \frac{1}{n} L_2(n+1).$$

Thus

$$(n+1)L_2(n+1) = nL_2(n) + \frac{1}{n+1}.$$

The solution is

$$L_2(n) = \frac{1}{n} \left(1 + \frac{1}{2} + \cdots + \frac{1}{n} \right).$$

Consequently, the conditional mean rupture time is given by

$$\mathbb{E}(\tau|R=n) = \frac{\int_0^\infty tp_n(t)dt}{\int_0^\infty p_n(t)dt} = \frac{1}{\sigma} \frac{L_2(n)}{L_1(n)} = \frac{1}{\sigma} \left(1 + \frac{1}{2} + \cdots + \frac{1}{n} \right).$$

1st April 2026